

times before wearing out. Conversely, sensors are still able to function properly if the skin is crumpled up. This makes them well-suited for use on the skin.

Multitasking skin

Researchers in Korea have developed their own compact electronic skin that multitasks like human skin, sensing pressure, temperature and sound (which is just air pressure, really) simultaneously.

The scientists took design cues from fingertips—home to some of the most sensitive skins on the human body. Fingerprint ridges help amplify vibrations, making sense of touch extremely fine-grained, so we can tell crinkly from grainy from velvety. Plus, the layers of dermis and epidermis underneath interlock with each other, providing more points of contact.

Colour-shifting electronic skin

The ability of animals like chameleons, octopus and squid to change their skin colour for camouflage, temperature control or communication is well known. While science has been able to replicate these abilities with artificial skin, the colour changes are often only visible to the naked eye when the material is put under huge mechanical strain.

Researchers from Tsinghua University have developed a new type of user-interactive electronic skin that can change colour. The changes are perceptible to the human eye without much level of strain. The skin could have applications in robotics, prosthetics and wearable technology.

Graphene, with its high transparency, rapid carrier transport, flexibility and large specific surface area, shows application potential for flexible electronics, including stretchable electrodes, super capacitors, sensors and optical devices. Researchers employed flexible electronics made from graphene, in the form of a highly-sensitive resistive strain sensor, combined with a stretchable organic



Chinese researchers' electronic skin can change colour like chameleons

electrochromic device. On exploring the substrate (underlying) effect on the electromechanical behaviour of graphene, they found subtle strain—between zero and 10 per cent—was enough to cause an obvious colour change, and the RGB value of the colour quantified the magnitude of the applied strain.

Powering the electronic skin

Stanford research team has developed a stretchable solar cell that could be used to power the electronic skin. An accordion-like micro-structure allowed the cells to stretch up to 30 per cent without damage. The team also added biological and chemical sensors to the skin to supplement pressure sensors.

Another research team has announced an electronic patch for monitoring patients' vital signs. Described as 'electric skin,' the device was created by embedding sensors in a thin film and then placing the film on a polyester backing similar to those found on temporary tattoos. A small coil provided power through induction. In tests, the device stayed in place for 24 hours without adhesives, relying instead on the Vander Waals force, and was flexible enough to move with the skin it was placed on.

Emerging applications

Electronic skins can monitor a person's heartbeat, brain activity, muscle contractions and more without the need for bulky conventional electronics. In addition to patient monitoring, these could be used to detect speech by sensing vibrations in the larynx

and emit heat to help in healing.

Electronic skin is no thicker than a human hair and can be applied as easily as a temporary tattoo. It sticks without the need for any glue, and can flex and stretch without breaking. In essence, electronic skin consists of mechanically flexible sensor networks that can wrap around

irregular surfaces, and spatially map and quantify various stimuli.

Long-term vision is to heterogeneously integrate a wide range of sensor networks (pressure, strain, light, temperature and humidity) and associated electronics on large-area plastic substrates using process schemes that are compatible with conventional silicon or LCD manufacturing lines. The sensor elements are based on a silicon, oxides, metals, piezoelectrics and/or organic materials that are processed at less than 35°C. The resulting electronic skin may find a wide range of applications in interactive input/control devices, smart wallpapers, robotics and medical/health monitoring devices.

Japanese researchers have taken a step closer to creating electronic skin by employing flexible electronics that can be worn as a second skin for biomedical and other applications. The aim is to make electronic skin as much a part of our daily lives as the clothes we wear.

If this happens, athletes will use electronic skin to view their heart rates, sugar levels and work rate. It could provide doctors with continuous data on patients' vital signs without the need for repeatedly attaching and removing medical equipment. The rest of us might employ the electronic skin to monitor body health metrics. Meanwhile, engineers could put down their tablets when doing tricky repair work and instead view maintenance procedures displayed on their arms. Such is the promise of ultra-thin, flexible and non-constraining electronic skin! **EFY**